

← Back

Week 2 Quiz

Graded Quiz • 30 min

Hide menu

Week 2 Lectures

Week 2 Quiz

Quiz: Week 2 Quiz

Submitted

🎉 Congratulations! You passed!

Grade received 100%

Latest Submission 100%

To pass 50% or higher

Go to next item

📝 Submit your assignment

To detect healthy vegetation the best wavelength range to use is (choose one)

Due Jun 30, 11:59 PM IST Attempts 3 every 2 hours

1 / 1 pointTry again

🎯 Receive grade

To Pass 50% or higher

Your grade 100%

View Feedback

We keep your highest score

👍 Like

👎 Dislike

📄 Report an issue

1. To detect healthy vegetation the best wavelength range to use is (choose one)

1 / 1 point

☐ visible green

☒ near infrared

☐ visible red

☒ Correct

Yes, that's correct. Green vegetation is strongly reflecting in the near infrared, and changes in condition also show up well.

2. To detect the difference between healthy green vegetation and the same vegetation that has lost its pigmentation due to senescence is to image in the (choose one answer)

1 / 1 point

☒ visible red

☐ visible green

☐ near infrared

☒ Correct

That is correct, because the loss of chlorophyll that accompanies plant drying is best seen in the visible red, where chlorophyll absorption no longer occurs.

3. To detect moisture stress in vegetation, and the drying of soils, imaging at which of the following wavelengths would be preferable?

1 / 1 point

☐ 0.5µm

☒ 1.45µm

☐ 0.75µm

☒ Correct

That is correct. It is in one of the water absorption bands.

4. Mismatched detector characteristics in a push broom scanner will lead to (choose one answer)

1 / 1 point

☒ line striping along the swath

☐ line striping across the swath

☐ both

☒ Correct

That is correct.

5. With regard to possible radiometric distortion over the range of visible and reflected infrared wavelengths, the solar radiation curve, described by Planck's law (choose one answer from below)

1 / 1 point

☐ is not important, in general

☐ is only important for hyperspectral sensors

☒ must be compensated for

☒ Correct

That is correct, otherwise the relative pixel values from band to band will not be the same as the reflectance of the earth surface cover types; they will be biased by the shape of the solar curve.

6. The wavelength dependence of atmospheric scattering is strongest when (choose the correct answer)

1 / 1 point

☐ the atmosphere is foggy

☒ the atmosphere is clear

☐ when there is light haze with scattered clouds

☒ Correct

That is correct. We then have Rayleigh scattering which shows the strongest dependence on wavelength

7. In the range of wavelengths commonly used in remote sensing the most significant absorbing constituent in the atmosphere is (choose the correct one)

1 / 1 point

☐ ozone

☐ carbon dioxide

☒ water vapour

☒ Correct

That is correct.

8. Panoramic (geometric) distortion is caused by (choose one of the following)

1 / 1 point

☐ the change in IFOV across the swath

☐ the platform being too close to the surface

☒ the difference in projected area on the ground across the swath at constant IFOV

☒ Correct

That is correct. At a constant angular IFOV the "pixel size" on the ground is larger at the swath edges than at nadir.

9. At its simplest level, correcting errors in image geometry using matrix mathematical models requires (choose one of the following)

1 / 1 point

☒ the distortions to be a linear function of the imaging coordinates

☐ the distortions to be no more complicated that a second order dependence on the imaging coordinates

☐ there is no restriction on the order of the distortion

☒ Correct

That is correct, because the matrix elements are the coefficient in a linear transformation.

10. Geometric distortion caused by earth rotation will be present for most imaging sensors in spacecraft remote sensing. Choose the correct answer.

1 / 1 point

☒ true

☐ false

☐ depends on the platform

☒ Correct

That is correct. Because the earth rotates at a finite rate and the spacecraft acquires an image over a finite time duration earth rotational distortion will always occur.

1/1

https://www.coursera.org/learn/remote-sensing/exam/MCOJ/week-2-quiz/attempt/redirectToCoverPage